LAB EXPERIMENT RECORD

20-10-2026---Set 23: Employee Performance Analysis

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Date	20 OCTOBER 2026
Day	Tuesday

Aim

To analyze employee performance data in R and Tableau using a bar chart, line chart, table and an interactive dashboard.

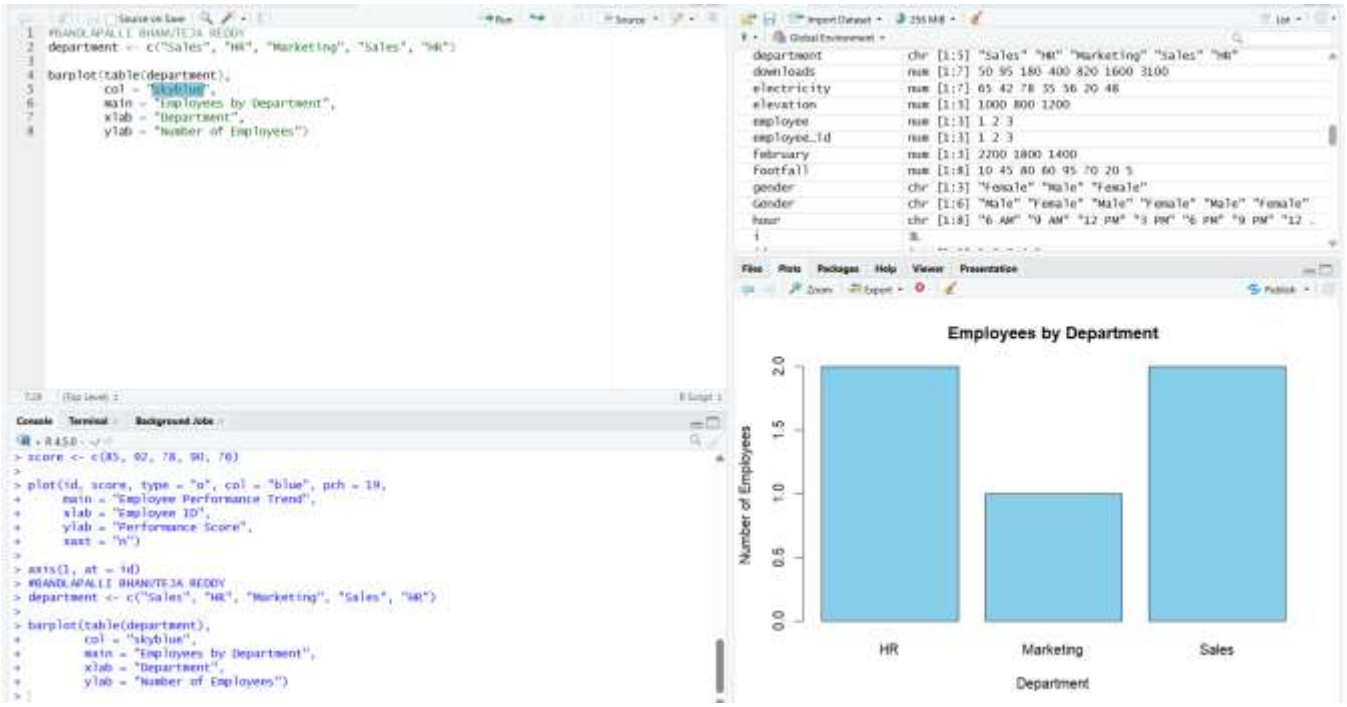
Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

Question 1: Bar Chart of Employees by Department

Question: In R, create a bar chart to visualize the distribution of employees across different departments. Label the chart elements.

Aim: To show the distribution of employees across departments using a labelled bar chart.

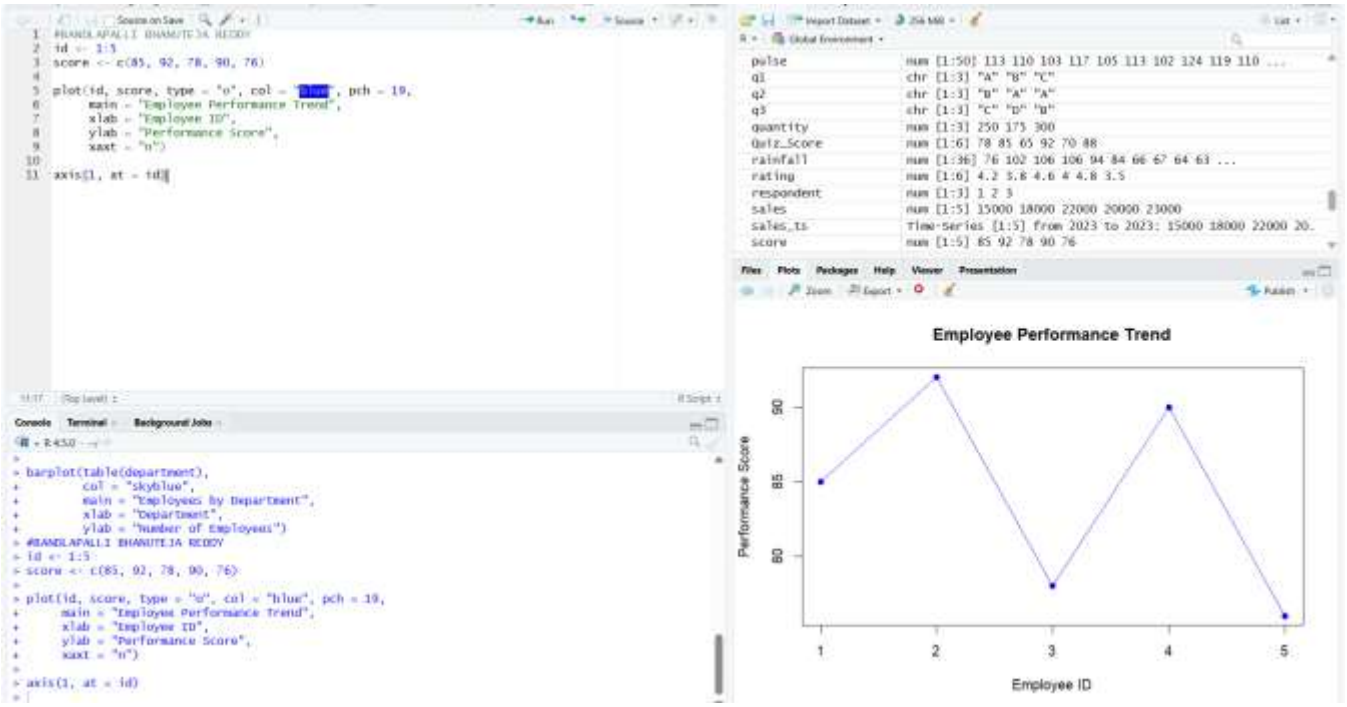


Result: Sales and HR have 2 employees each and Marketing has 1 employee.

Question 2: Line Chart of Performance Trend

Question: In R, generate a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

Aim: To visualize the performance trend of employees using a line chart with labelled axes and a title.

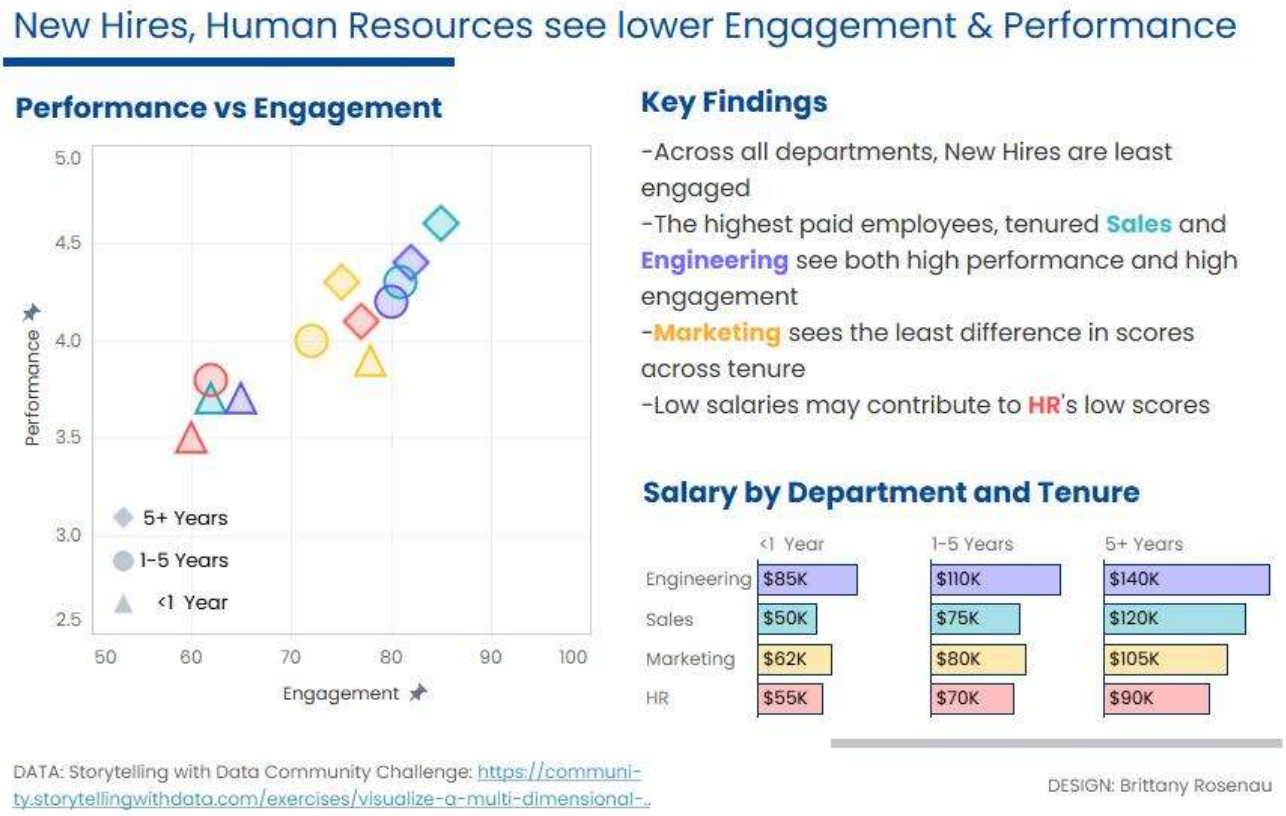


Result: Performance goes 85, 92, 78, 90 and 76, so it peaks at 92 for Employee 2 and is lowest at 76 for Employee 5, with no steady trend.

Question 3: Tableau Dashboard of Departments and Years of Service vs Performance

Question: In Tableau, build a dashboard combining a bar chart showing department distribution and a scatter plot displaying the relationship between years of service and performance scores.

Aim: To build an interactive Tableau dashboard that combines the department bar chart and the years-of-service scatter plot.



Result: The bar chart shows Sales 2, HR 2 and Marketing 1, and the scatter plot shows a very weak negative correlation of about -0.17, so years of service does not clearly predict performance.

Question 4: Table of Employee Performance

Question: In R, develop a table showing the performance data for each employee. Label the table elements.

Aim: To present the department, years of service and performance score of each employee in a labelled table.

```
#BANDLAPALLI BHANUTEJA REDDY
data <- data.frame(
  Employee_ID = 1:5,
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),
  Years_of_Service = c(5, 3, 7, 4, 2),
  Performance_Score = c(85, 92, 78, 90, 76)
)

print(data)
```

```
+         xlab = "Department",
+         ylab = "Number of Employees")
> #BANDLAPALLI BHANUTEJA REDDY
> data <- data.frame(
+   Employee_ID = 1:5,
+   Department = c("Sales", "HR", "Marketing", "Sales", "HR"),
+   Years_of_Service = c(5, 3, 7, 4, 2),
+   Performance_Score = c(85, 92, 78, 90, 76)
+ )
>
> print(data)
  Employee_ID Department Years_of_Service Performance_Score
1           1      Sales                5                85
2           2         HR                3                92
3           3  Marketing                7                78
4           4      Sales                4                90
5           5         HR                2                76
>
```

Result: The table lists all five employees; the department averages are Sales 87.5, HR 84 and Marketing 78.

Overall Result

Observation: The average performance score is 84.2, the average service is 4.2 years, and the highest score (92) belongs to Employee 2 from HR.